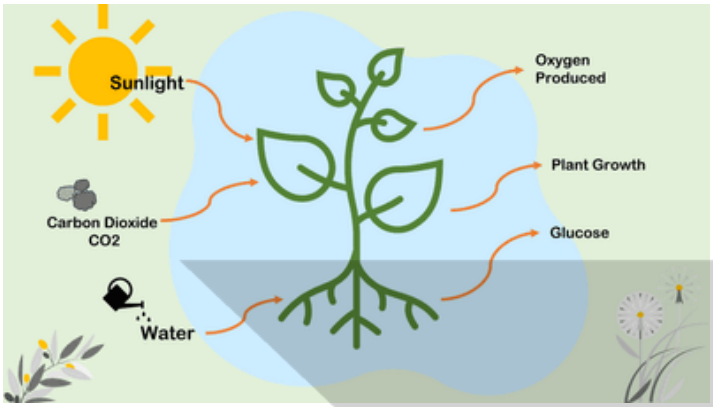
# 1. Plant Nutrition

## 1.1. Photosynthesis

**Photosynthesis:** the process by which plants manufacture carbohydrates from raw materials using energy from light.




$$Carbon Dioxide + Water \xrightarrow{light+chlorophyll} Glucose + Oxygen$$

$$6CO_2 + 6H_2O \xrightarrow{light+chlorophyll} C_6H_{12}O_6 + 6O_2$$

- Only **EXTENDED Students** must know the **Balanced Chemical Equation**
- The carbon dioxide diffuses through the open stomata of a plant leaf, and water is taken up through the roots.
- Chlorophyll is a **green pigment** that traps light energy and converts it into chemical energy to form carbohydrates and their subsequent storage.
- Glucose is used for respiration, energy storage, cellulose cell walls, and making proteins and sugars.

**Use and Storage of the Carbohydrates Made in Photosynthesis**

- starch as an energy store
- cellulose to build cell walls
- glucose used in respiration to provide energy
- sucrose for transport in the phloem

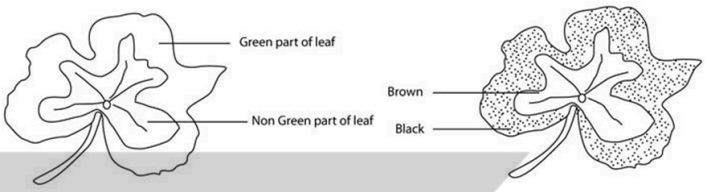
## 1.2. Mineral Requirements

| Nitrate ions | Magnesium ions |
|---|---|
| Making amino acids | Making chlorophyll |
| Deficiency: small plant due to slow/stunted growth | Deficiency: The plant lacks chlorophyll, leaves turn yellow. |

**Tip!** You need to know the purpose of these required nutrients.

## 1.3. Investigation of Chlorophyll

- Take a potted plant with variegated (green and white) leaves.
- De-starch the plant by keeping it in complete darkness for about 48 hours.
- Expose the plant to sunlight for a few days.
- Leaf boiled in water for 2 minutes to break down cell walls, denature enzymes and allow for easier penetration by ethanol.
- Warmed in ethanol until the leaf is colourless to extract chlorophyll, which would mask the observation
- Dipped into the water briefly: to help soften the leaf
- The leaf is placed on a white tile, and iodine is added. If starch is present, the colour will be blue-black; if absent, it will remain brown.

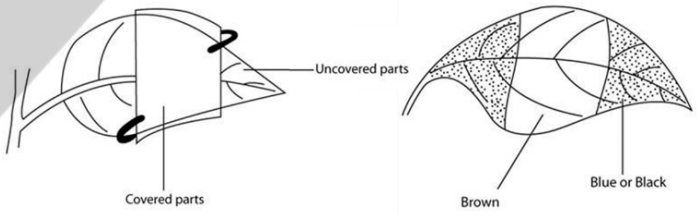



(a) variegated leaf before (b) after testing for starch

## 1.4. Investigation of Light Intensity

**NOTE: This type of question is frequently asked in Paper 2.**

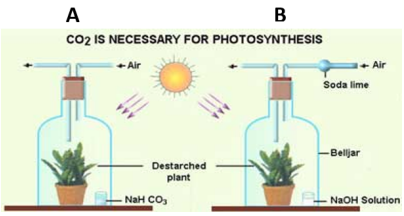
- De-starch the plant by keeping it in the dark for 48 hours
- Place a stencil over part of a leaf
- Place the leaf in sunlight for 4-6 hours
- Remove the stencil and test for starch
- **+ve result = parts which received light turn blue-black**
- **-ve result = parts that didn't receive light remain brown**




(a) leaf covered with stencil (b) after testing for starch

## 1.5. Investigation of Carbon Dioxide Concentration

- Take two de-starched potted plants.
- Cover both the plants with bell jars and label them A and B.
- Inside A, keep $NaHCO_3$ (Sodium Bicarbonate). It produces $CO_2$ .
- Inside B, keep $NaOH$ (Sodium Hydroxide). It absorbs $CO_2$ .
- Keep both set-ups and do the starch test in the sunlight for at least 6 hours

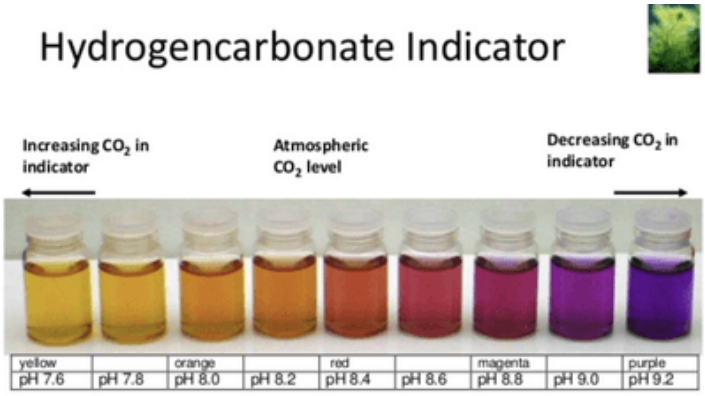



- The leaves of Plant A will turn black after the starch test
- The leaves of Plant B will remain brown after the starch test



Copyright © 2025 ZNotes Education & Foundation. All Rights Reserved.
This document is authorised for personal use only by Faiq Ali at Enderun Colleges on 05/04/26.

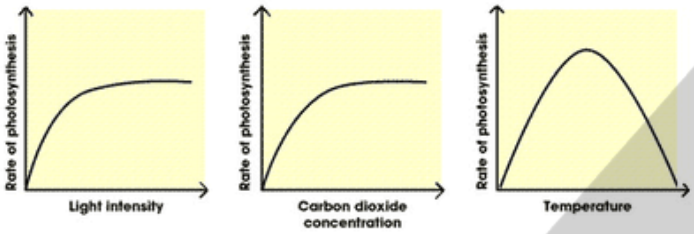
**Hydrogencarbonate indicator**: measures the carbon dioxide concentration



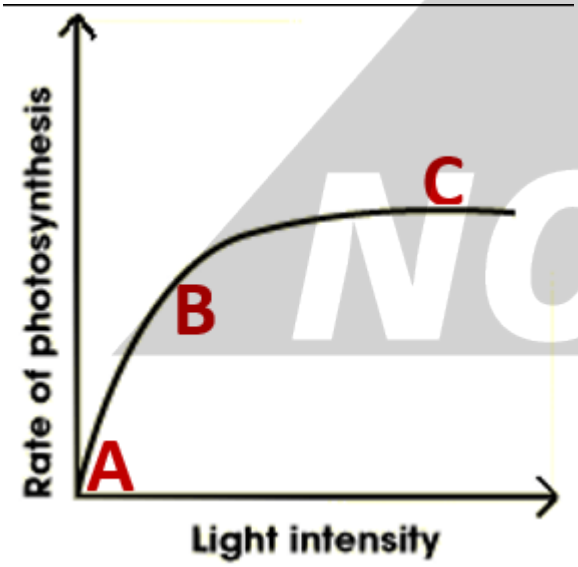

**Key:** PMROY, purple being the lowest carbon dioxide concentration, and yellow the highest.

## 1.6. Limiting Factors

**Limiting Factors**: Conditions where something present in the environment (Temperature, Carbon Dioxide Concentration and Light Intensity) is in such short supply that it restricts life processes.



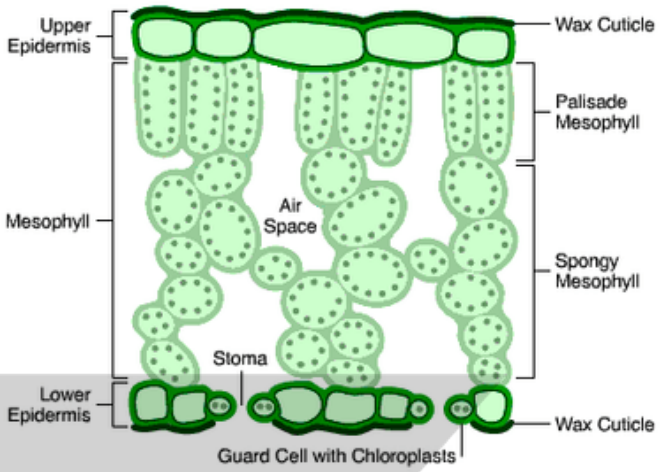

### Light intensity




- As the amount of light increases, the rate of photosynthesis increases (A-B)
- The limiting factor is light
- Increasing the amount of light after a certain point does not affect the rate (C)
- The limiting factor is now carbon dioxide or temperature.

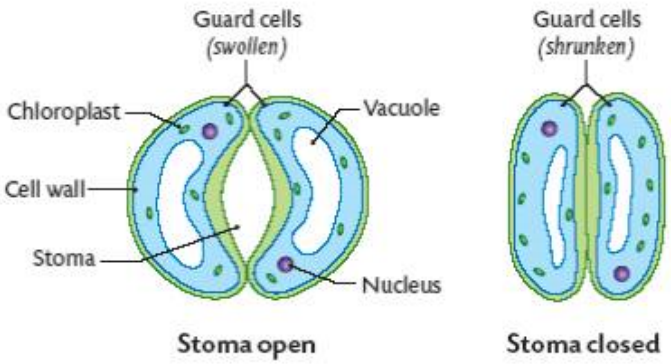
## 1.7. Leaf Structure

Most dicotyledonous plant leaves have a **large surface area** and are **thin**.




| Structure | Function |
|---|---|
| Waxy Cuticle | A waterproof waxy layer that prevents water loss from the top of the leaf |
| Upper & Lower Epidermis | Thin and transparent to allow light to enter the palisade mesophyll cells |
| Palisade Mesophyll Cells | Found at the top of the cell and contains the **MOST chloroplasts** that absorb sunlight for photosynthesis. |
| Spongy Mesophyll Cells | Irregularly shaped cells create air spaces to allow the gaseous exchange to take place; do not contain many chloroplasts |
| Vascular Bundle | Made up of xylem and phloem |
| Xylem | A tissue vessel that transports water and mineral ions |
| Phloem | A tissue vessel that transports sucrose and amino acids |
| Stomata/Stoma | Tiny gaps that open and close to allow the gaseous exchange to occur |
| Guard Cells | Control the opening and closing of stomata |

### Opening and closing of Stomata:






Copyright © 2025 ZNotes Education & Foundation. All Rights Reserved.
This document is authorised for personal use only by Faiq Ali at Enderun Colleges on 05/04/26.

- Stomata open when guard cells absorb water by osmosis, become turgid, and curve outward, creating a gap.
- They close when guard cells lose water, become flaccid, and collapse, sealing the pore to reduce water loss.

Syllabus 6.2.3: You must be able to explain how the structures above adapt leaves for photosynthesis

## 1.8. Adaptations of Leaf Structure for Photosynthesis

- Here is a table of different adaptations of the specific leaf structure helping towards photosynthesis.

| Adaptation | Functions |
|---|---|
| Large Surface Area of Leaf | Increase surface area for diffusion of carbon dioxide and absorption of light for photosynthesis. |
| Thin | Allow carbon dioxide to diffuse quickly into the palisade mesophyll cells |
| Chlorophyll | Absorb light energy so photosynthesis can take place. |
| Network of Veins | Allow efficient transport of water and mineral ions throughout the plant |
| Epidermis is thin | Allow more light to reach the palisade mesophyll cells |
| Stomata/Stoma | Allow oxygen and carbon dioxide to continuously diffuse in and out of the leaf |



Copyright © 2025 ZNotes Education & Foundation. All Rights Reserved.
This document is authorised for personal use only by Faiq Ali at Enderun Colleges on 05/04/26.